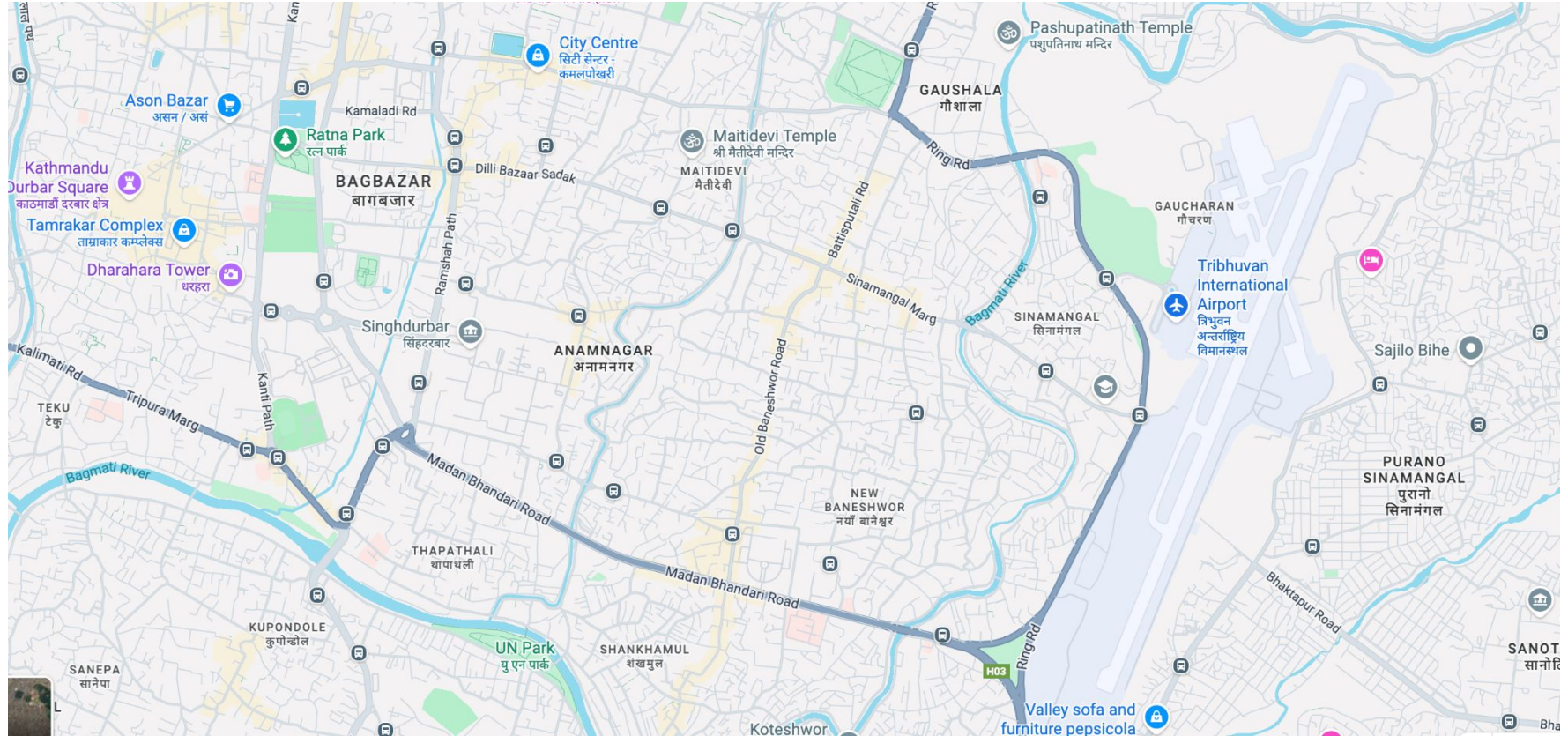




RVA, Qualitative, and Quantitative Assessment Of a Residential Building

Vulnerability Assessment & Retrofitting Techniques Assignment Presentation
IoE Pulchowk Campus | Jan 2026

Region of Study



RVA & Qualitative Assessment

Seismicity and Target Performance Level



Location: Baneshwor, Ward number 10, KMC

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Soil Type D (Table 4-3: Sub-soil Type D sites inside Kathmandu Valley NBC 105:2025) [1]

PGA 0.35 (Annex C: Seismic Zoning Factor For Cities/ Municipalities NBC 105:2025) [1]

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Building Type: Residential building

Life Safety* - reducing the risk of life loss in the largest expected earthquake. [3]

*Buildings meeting the Life Safety performance level are expected to experience little damage from relatively frequent, moderate earthquakes. But significantly more damage and potential economic loss from the most severe and infrequent earthquakes could affect them. (DUDBC & UNDP: Seismic Vulnerability Evaluation Guideline For Private and Public Buildings, May 2011) [3]

Selected RC Building

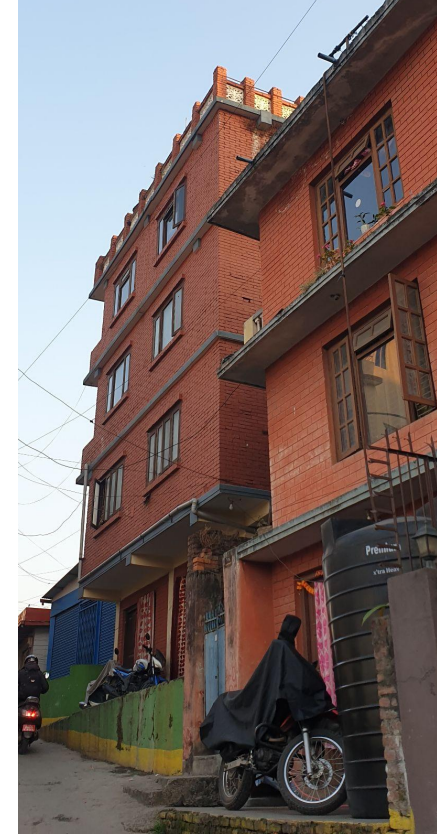
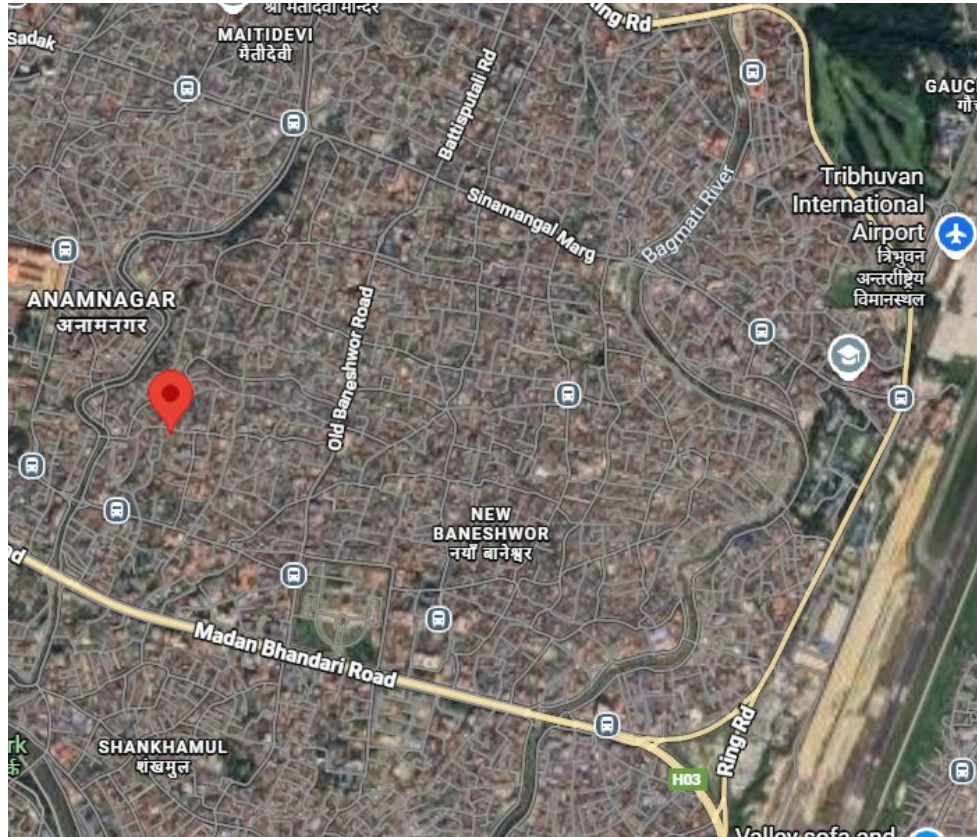


Table 1: Criteria for building being unsafe [3]

S.N.	Conditions	Posting
1	Building has collapsed, partially collapsed, or moved off its foundation	Unsafe
2	Building or any story is significantly out of plumb	Unsafe
3	Obvious severe damage to primary structural members, severe cracking of walls, severe cracking of columns, beam-column joints, buckling of reinforcement bars, or other signs of severe distress present	Unsafe
4	Obvious parapet, chimney, or other falling hazard present	Area Unsafe
5	Large fissures in ground, massive ground movement, or slope displacement present	Unsafe
6	Other hazard present (e.g. fallen power line, fallen tree)	Unsafe or Area Unsafe

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Since none of the conditions persist, the building was safe for interior and exterior inspection.

Building information (obtained via Site Visit),

- Age: 44 years
- Bay Width: 12ft, 12ft
- Storey Height: 10ft
- No of bays: 1, 2
- No of Storey: 4
- Material: M20
- Steel: Fe 415
- Beam Reinforcement: 3-12 Φ + 2-12 Φ
- Beam Size: 13" * 9"
- Column Reinforcement: 8-12 Φ
- Column Size: 9" * 9"
- Slab thickness: 4"
- No of residents: 22
- Reinforced concrete frame with unreinforced brick masonry infill in cement mortar (115mm)
- not structurally designed and supervised by engineers during construction.

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Type 3: Non-engineered Reinforced Concrete
Moment Resisting-Frame Buildings (Table 1 [3])

Table 2 (d) Non-Engineered Reinforced Concrete Frame Buildings (≥ 4 storey) [3]

Shaking Intensity (MMI)		VI	VII	VIII	IX
PGA (%g)		5-10	10-20	20-35	>35
Damage Grade for different classes of buildings	Weak	DG1	DG2	DG4	DG5
	Average	-	DG1	DG3	DG4
	Good	-	DG1	DG2	DG3

Structural Assessment Checklist (Annex III) [3]



Building System			
Load Path	C	Vertical Discontinuities	C
Redundancy	C	Mass	NK
Geometry	C	Torsion	NK
Mezzanines/Loft/Subfloors	N/A	Adjacent Buildings	C
Weak Story	NC	Flat Slab Frames	C
Deterioration Of Concrete	C	Short Columns	C
Crack In Boundary Columns	C	Masonry Units	NC
Soft Story	NC	Masonry Joints	C
Crack In Infill Walls	C		

Structural Assessment Checklist (Annex III) [3]

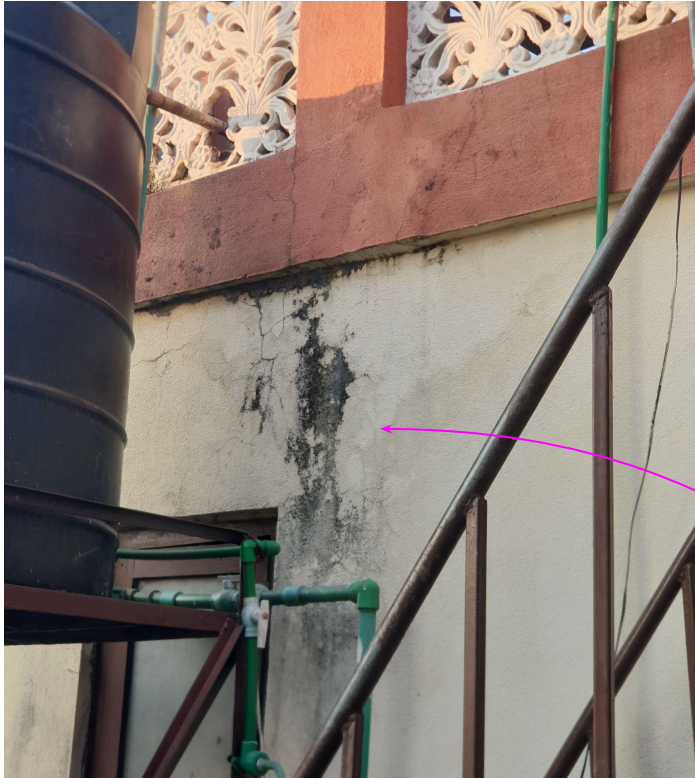


Large Openings
at Ground Floor,
Walls over the
projected slab



Visual
Deterioration

Structural Assessment Checklist (Annex III) [3]



Visual
Deterioration

Structural Assessment Checklist (Annex III) [3]



Lateral Load Resisting System			
Shear Stress In Rc Frame Columns	NK	Beam Bar Splices	NK
Shear Stress In Shear Walls	N/A	Column Tie Spacing	NK
Shear Stress Check For Rc Masonry Infill Walls	NK	Stirrup Spacing	NK
Axial Stress In Moment Frames	NK	Joint Reinforcing	NK
No Shear Failures	NK	Stirrup And Tie Hooks	NK
Concrete Columns:	C	Joint Eccentricity	NK
Strong Column/Weak Beam	NK	Wall Connections	N/A
Beam Bars	C	Interfering Walls	C
Columns Bar Splices	NK		

Structural Assessment Checklist (Annex III) [3]



Diaphragms		Geologic Site	
Diaphragm Continuity	C	Area History	N/A
Plan Irregularities	NK	Liquefaction	N/A
Diaphragm Reinforcement At Openings	N/A	Slope Failure	N/A

Damage Grade and Posting Classification



Grade 2: Moderate damage (Annex IV [3])

Structural damage : **Slight**

Non-structural damage: **Moderate**

Table 2: Posting Classifications [3]

Posting Classification	Color	Description
INSPECTED	Green	No apparent hazard found, although repairs may be required. Original lateral load capacity not significantly decreased. No restriction on use or occupancy



INSPECTED

NO RESTRICTIONS ON USE OR OCCUPANCY

This structure has been inspected (as indicated below) and no apparent structural hazard has been found. Report any unsafe conditions to local authorities; reinspection may be required.

- ☐ Exterior Only
☒ Exterior and Interior

Facility Name and Address:

Residential Building, Thapagaun,
Baneshwor

Date 14 Jan. 2026
Time 5:00 pm

This facility was inspected under emergency conditions for:

Rapid Visual Assessment
(Jurisdiction)
on the date and time noted.

Inspector ID/Agency:

0786ced94, IoE Pulchowk Campus

Do Not Remove This Placard until
Authorized by Governing Authority.

Quantitative Assessment

Conclusion and Recommendations



The safety of the building is moderate, thus retrofit is necessary.

The building needs some immediate repair and regular maintenance.

Further structural analysis is recommended to determine extent of retrofit, to ensure adequate performance during future earthquakes.

- [1] Department of Urban Development and Building Construction (DUDBC), Seismic Design of Buildings in Nepal (Second Revision), NBC 105: 2025, Kathmandu, Nepal: Ministry of Urban Development, 2025.
- [2] Federal Emergency Management Agency (FEMA), Rapid Visual Screening of Buildings for Potential Seismic Hazards: A Handbook, FEMA-154 Data Collection Form, 3rd ed., Washington, DC, USA, 2015.
- [3] Department of Urban Development and Building Construction (DUDBC), Seismic Vulnerability Evaluation Guideline for Private and Public Buildings, Earthquake Risk Reduction and Recovery Preparedness Programme for Nepal, Kathmandu, Nepal: Ministry of Physical Planning and Works, 2011.

Discussions

Shot on Mi A2
Compiled on Google Slides

Thank you for your Time

Presenter

Niraj Kumar Yadav
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Level : BCE IV/II

Submitted To

Prof. Kshitij C. Shrestha
Department of Civil Engineering
IoE Pulchowk Campus